

Luminous Dynamics
Richardson, TX

February 10, 2026

Editorial Board

Dear Editors,

We are pleased to submit our manuscript, “**Hyperdimensional Active Inference for Consciousness-Aware Distributed Intelligence**,” for your consideration.

Summary. This paper introduces Symthaea, a novel architecture that unifies four independently validated paradigms—Hyperdimensional Computing (HDC), Integrated Information Theory (IIT), Liquid Time-Constant Networks (LTC/CfC), and Active Inference—into a single coherent framework for consciousness-aware distributed intelligence. To our knowledge, this is the first system to combine all four paradigms, placing it in a unique “blue ocean” with no direct competitors.

Key contributions:

1. **Unified architecture** integrating HDC (16,384-dimensional holographic vectors), IIT (Φ proxy with Spearman $\rho = 0.50$ against exact computation), LTC/CfC (real-time neural dynamics at 59K–87K inferences/sec), and Active Inference ($7.9\times$ speedup over pymdp).
2. **Consciousness-aware federated learning** with Byzantine fault tolerance validated to 34%, epistemic-weighted aggregation, and differential privacy with Rényi accounting.
3. **16 validated benchmarks** across neuroscience, ethics, speech, vision, and distributed systems, including 92.9% on the ETHICS benchmark (HDC-only, no LLM), 94.5% speaker identification, and biologically-grounded *C. elegans* connectome analysis (448 neurons, 7,379 connections).
4. **Open-source implementation** in Rust (~ 338 K LOC, 3,200+ tests), demonstrating that consciousness-inspired computation is both theoretically grounded and practically viable.

Novelty. While Liquid AI (LFM2/LFM2.5) validates LTC commercially and FedHDC demonstrates federated HDC, no existing system combines all four paradigms. Our HDC moral algebra achieves 92.9% on ETHICS without any language model, using only 16,384-dimensional vectors with compositional operators—a fundamentally different approach to ethical reasoning.

Reproducibility. All benchmarks are reproducible via standalone Rust examples. Results files, supplementary reproduction instructions, and the complete codebase are included.

We believe this work is well-suited for [Journal Name] given its interdisciplinary scope spanning

computational neuroscience, distributed systems, and AI safety.

Sincerely,

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